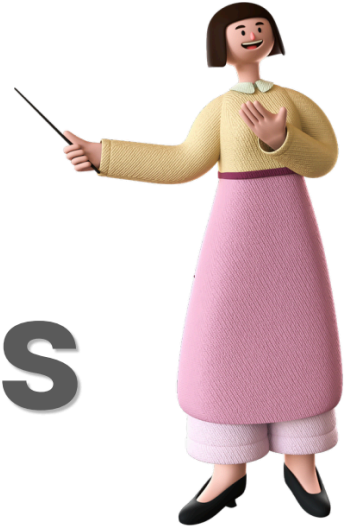




Control Flow Statements



“Every problem you solve today is one step closer to your dream company.”

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WHAT ARE CONTROL FLOW STATEMENTS?



Control flow statements decide how a program executes step-by-step.

By default, a program runs line by line (top → bottom)

Control flow statements allow you to change this flow

Simple real-life example:

“If it rains → take umbrella

Else → go normally”

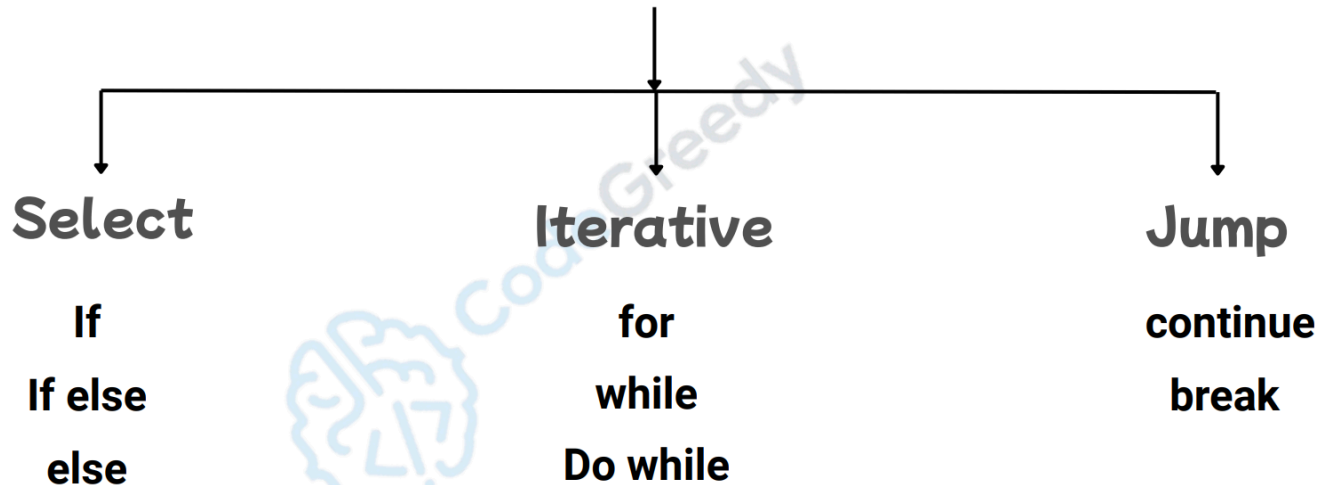
This is exactly how programs make decisions.



TYPES OF CONTROL FLOW STATEMENTS



Control Statements



if and else if



```
if (expression){
```

```
    statements
```

```
}
```

```
else{
```

```
}
```

Any Non-Zero value
(+ve, -ve)



Any of these two, only one block will get executed.

if (condition) → if (expression)

```
}  
else {  
{
```

```
}  
else {  
{
```

```
}
```

any statement
that has
value

```
if (4)  
{ print("Hello")  
}  
else  
{ print("Bye")  
}
```

int x=10, y=20

```
if (x <= y)
{
    _____
    _____
    _____
}
else
{
    _____
    _____
    _____
}
```

```
if (x <= y)
{
    print("Hello");
    print("Bye");
    print("Get");
}
```

```
int x = 10, y = 20
```

```
} Hello GN
```

```
if (x ≤ y)  
    print("Hello");
```

```
else  
    print("Bye");  
    print("GN");
```

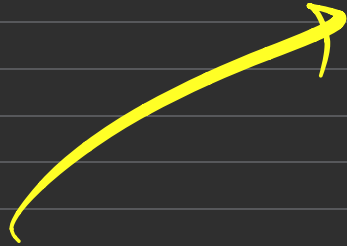
```
public class Main {  
    public static void main(String[] args) {  
  
        int x = 10, y = 20;  
  
        if (x <= y)  
            System.out.println("Hello Pramod");;  
    }  
}
```

Empty lines



```
if ( 4 ≤ 100 )  
    print ( "Hello" );  
    print ( " Bye" );
```

```
else  
    print ( " 4N" );  
    print ( " 4M" );
```

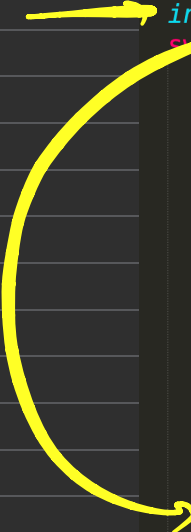


~~Error~~

Switch

```
switch (expression)
{
    case 1:
        ==
        break
    case 2:
        ==
        break
    default:
        ==
        break
}
```

$x = 3$



```
int x = 3;
switch(x){

    case 1:{
        System.out.println("hello");
        break;
    }

    System.out.println("I'm outside of the case");

    case 2:{
        System.out.println("Good Bye");
        break;
    }

    case 3:{
        System.out.println("Morning");
        break;
    }

    default:{
        System.out.println("Bye");
        break;
    }

}
```

9 ~~a~~ ~~x~~ A → C prog
10 ~~b~~ 2 ~~B~~ → Java
11 ~~c~~ ~~z~~ C → Python
12 ~~d~~ ~~y~~ D → NOT

(29)th

~~(30)~~

~~28~~

~~20~~

15

Loop

Mainly used to do repeated work until certain condition

for

while

Do while



for loop

```
for (init; cond; incr) dec;
```

```
{
```

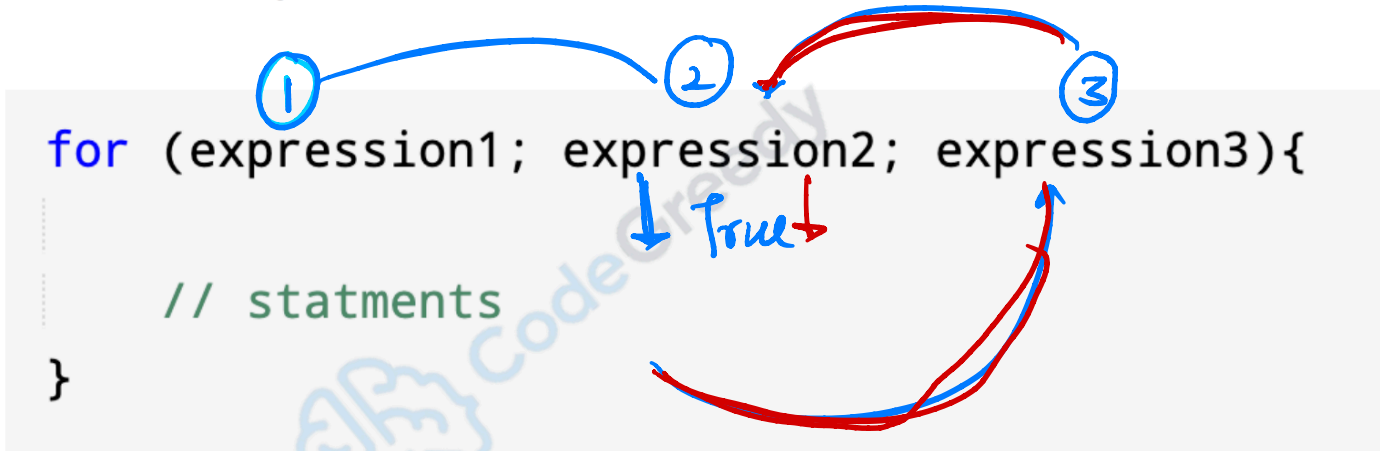
```
for (1; i < 3)  
    print(i)
```

```
for (1; 2; 3)  
    print("hello");
```

```
for ( ; ; )  
    print("hello")
```

for Loop

for loop is mainly used when we know number of iterations.



All the expression in the for loop is optional - it means it's okay if you won't write it.

while Loop

for loop is mainly used when we don't know number of iterations - but know certain condition.

```
while (expression1){  
  
    // statments  
  
}
```

expression in the while loop is compulsory to have it.

int i;

for (i=0; i ≤ 10; i++)

{ print("Hello");

}



int i=0

while (i ≤ 10)

{ print("Hello")

i++

}

do while Loop

It mainly executes and then checks the conditions.

```
do{  
  
    // statments  
  
}while (expression);
```

expression in the do while loop is compulsory to have it.

```
int x=10
```

```
do
```

```
{
```

```
    print("Hello")
```

```
    x++
```

```
} while ( x > 100 );
```

break

break statement is mainly used to take you outside of the current loop.

break can be used only with the loops - it cannot be used with any other statement.

for, while, do while

continue

Mainly skips the current iteration and take you to the next Iteration.

continue can be used only with the loops - it cannot be used with any other statement.

```
for (int i = 1; i <= 10; i++){  
    if (i == 5){  
        break;  
    }  
    print(i);  
}
```



Output

1 2 3 4

The moment `i == 5` `break` will take you out of the loop. Hence, loop terminates.

~~i = 2~~

j =

for (i = 1; i ≤ 3; i++)

{

for (j = 1; j ≤ 3; j++)

{

if (j = 2) break

print(j)

,
}

}

```
for (int i = 1; i <= 10; i++){  
    if (i % 2 == 0){  
        continue;  
    }  
    print(i);  
}
```

Handwritten notes:
- $4 \div 2 = 0$ (above the if condition)
- A red box around `continue;` with a blue arrow pointing to the next iteration.
- skip (written in blue below the if block)

Handwritten: $i = 1, 2, 3, 4$ (with 2 and 4 crossed out)
CodeGreedy

Output

1 3 5 7 9

Handwritten: 2, 4, 6, 8

When i is even - continue means it skips the below statements and move to the next iterations.

Char c = 'b'

(
Char c = 'g'

binary
code